TSM 099 Polynomials Test - Practice 2

Version 202610122020.20 | Name: _____

Complete and show all work on a separate sheet of paper. Record final answers on the answer sheet. Give exact answers and simplify completely.

Answer key follows the answer sheet. Attempt the test before checking it.

1. Factor completely.

$(a)\quad x^2 + 2x - 15$

$(b)\quad 4x^2 - 9$

2. Factor completely over the integers, or write prime.

$(a)\quad 3x^2 + 2x + 1$

$(b)\quad 3x^2 + x - 2$

3. Factor out the greatest common factor.

$6x^4 - 3x^3 + 12x^2$

4. Multiply and simplify.

$(3x - 3)(4x + 2)$

5. Factor using the ac-method and grouping. Show ac, the split middle term, and the grouping.

$12x^2 + 25x + 12$

Student answer sheet

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Question 1

(a) 10 pts

(b) 10 pts

Question 2

(a) 10 pts

(b) 10 pts

Question 3

Factored form 20 pts

Question 4

Polynomial 20 pts

Question 5

ac; pair 4 pts

Split middle term 4 pts

Grouping 6 pts

Factored form 6 pts

Answer key

TSM 099 Polynomials Test | Practice 2

Version 202610122020.20 | Name: _____

Question 1

(a)	$(x-3)(x+5)$	10 pts
(b)	$(2x-3)(2x+3)$	10 pts

Question 2

(a)	Prime	10 pts
(b)	$(x+1)(3x-2)$	10 pts

Question 3

Factored form	$3x^2(2x^2-x+4)$	20 pts

Question 4

Polynomial	$12x^2-6x-6$	20 pts

Question 5

ac; pair	$ac=144;\ 9,\ 16$	4 pts
Split middle term	$12x^2+9x+16x+12$	4 pts
Grouping	$3x(4x+3)+4(4x+3)$	6 pts
Factored form	$(3x+4)(4x+3)$	6 pts